

Homework for Chap 13

Given a direct bandgap semiconductor GaAs with a bandgap energy $E_g = 1.42$ eV at room temperature (300 K), its absorption coefficient satisfies the relationship with the incident photon energy E : when $E \geq E_g$, $\alpha \propto (E - E_g)^{1/2}$. When $E < E_g$, $\alpha \approx 10$ cm⁻¹ (negligible), here E is the photon energy. A monochromatic light is normally incident on a thin GaAs slab with the following conditions:

- Incident light wavelength $\lambda_1 = 1000$ nm;
- Incident light wavelength $\lambda_2 = 800$ nm;

Suppose the thickness of the GaAs slab $d = 1$ μ m, the Reflectivity of the semiconductor surface $R = 0.3$ for both λ_1 and λ_2 , and neglect multiple reflections:

1. Determine whether the incident light with wavelengths λ_1 and λ_2 can be effectively absorbed by the GaAs semiconductor, respectively, and explain the reasons.
2. If the absorption coefficient corresponding to the incident light of wavelength λ_1 and λ_2 is $\alpha = 10$ cm⁻¹ and $\alpha = 1 \times 10^4$ cm⁻¹, respectively, calculate the transmittance T of the light after passing through the GaAs slab for light with λ_1 and λ_2 .
3. In a photoluminescence spectrum measurement, a Ti:sapphire continuous wave (cw) laser delivers monochromatic light with wavelength $\lambda_2 = 800$ nm and the laser output power is 10 mW. If the laser beam is focused uniformly onto a round area with a radius $r = 100$ μ m. Given the $R = 0.3$, $\alpha = 1 \times 10^4$ cm⁻¹, the photogenerated carrier lifetime $\tau = 1$ ns, and suppose the photogenerated carriers are confined in the cylinder with the height $h = 1$ μ m, and the radius of top base and bottom base $r = 100$ μ m, calculate the average density of the photogenerated electrons.